Basis and Dimension

Consider the following examples which use the ideas of span, linear dependence and independence.

Example 1. The set of vectors $(1,1,0),(0,-1,1)$ and $(1,0,1)$ **does not span** $\mathbb{R}^3$ since the vector $(1,0,0)$ cannot be expressed as a linear combination of the given vectors. This can be shown as follows. If we write $(1,0,0) = \alpha_1(1,1,0) + \alpha_2(0,-1,1) + \alpha_3(1,0,1)$ and try to solve the resulting system

$$\alpha_1 + \alpha_3 = 1; \quad \alpha_1 - \alpha_2 = 0; \quad \alpha_2 + \alpha_3 = 0$$

we will get no solution.

Example 2. The set of vectors $(1,0,0),(0,1,1),(1,0,1),(0,1,0)$ spans $\mathbb{R}^3$. We prove this by showing that there exists $\alpha_i's$ such that any vector $(a,b,c) \in \mathbb{R}^3$ can be written as

$$(a,b,c) = \alpha_1(1,0,0) + \alpha_2(0,1,1) + \alpha_3(1,0,1) + \alpha_4(0,1,0)$$

However, $(0,1,0)$ depends on the first three vectors as $(0,1,0) = (1,0,0)+(0,1,1)-(1,0,1)$. In such a case, all the four vectors are **not necessary** since

$$\begin{aligned}(a,b,c) &= \alpha_1(1,0,0) + \alpha_2(0,1,1) + \alpha_3(1,0,1) + \alpha_4(0,1,0)\\ &= \alpha_1(1,0,0) + \alpha_2(0,1,1) + \alpha_3(1,0,1) + \alpha_4[(1,0,0) + (0,1,1) - (1,0,1)]\\ &= (\alpha_1+\alpha_4)(1,0,0) + (\alpha_2+\alpha_4)(0,1,1) + (\alpha_3-\alpha_4)(1,0,1)\end{aligned}$$

That is, even if four vectors are given, one of them is redundant which makes the set a linearly dependent set.

Another way to look at the above "*not necessary*" vectors is the following. We can write any vector $(a,b) \in \mathbb{R}^2$ as a linear combination in the following ways

$$\begin{aligned}(a,b) &= a(1,0) + b(0,1)\\ &= a(1,0) + b(0,1) + 0(1,1) = a(1,0) + b(0,1) + 0(1,1) + 0(2,1) = \cdots\end{aligned}$$

Thus the set $\mathcal{B} = \{(1,0),(0,1)\}$ is just enough (*not a vector less and not a vector more*) to generate any point in $\mathbb{R}^2$. Any other vector added to $\mathcal{B}$ does not really make any independent contribution to the linear combination.

One can also see that asking the maximum number of vectors that can span $\mathbb{R}^2$ does not make sense. As the example shows, (a,b) can be written as a linear combination of any number of vectors in $\mathbb{R}^2$. But the zero coefficients tell us that exactly two vectors $(1,0)$ and $(0,1)$ are required to span $\mathbb{R}^2$. (*Can you recall why just one vector is not enough to span* $\mathbb{R}^2$*, that is generate all the points* (a,b)*?*)

Thus, we are looking for a set $\mathcal{B} = \{u_1, u_2, \cdots, u_n\}$ that **contains no redundant vectors and any vector in the vector space or its subspace is a linear combination of the given vectors in** $\mathcal{B}$. Clearly, as the above example shows, such a set cannot be linearly dependent since a linearly dependent set will always contain at least one redundant vector. This gives us the concept of **basis** of a vector space which is defined as follows.

Definition of basis

Definition 1. Let V be a vector space. A basis $\mathcal{B}$ for V is a set of vectors in V such that

- $\mathcal{B}$ contains linearly independent vectors. (This property tells us that a basis does not contain any redundant vector.)
- span of the set $\mathcal{B}$ is equal to the entire vector space. (This property tells us any vector of the vector space can be written as a linear combination of the vectors in the set $\mathcal{B}$.)

Let us look at some examples. The idea is that while proving $\mathcal{B}$ is a basis, we need to prove that $\mathcal{B}$ satisfies both the defining properties.

Example 3. Verify if the set $\mathcal{B} = \{(1,1,1),(0,1,1),(0,0,1)\}$ of vectors in $\mathbb{R}^3$ is a basis for $\mathbb{R}^3$.

- First, we check that the vectors are linearly independent. Form the equation

$$\alpha_1(1,1,1) + \alpha_2(0,1,1) + \alpha_3(0,0,1) = (0,0,0)$$

  This gives $\alpha_1 = 0$, $\alpha_1 + \alpha_2 = 0$ and $\alpha_1 + \alpha_2 + \alpha_3 = 0$ which implies $\alpha_1 = \alpha_2 = \alpha_3 = 0$. Thus the vectors are linearly independent.

- Second, we check if the span of the set $\mathcal{B}$ is same as the vector space $\mathbb{R}^3$.

  Let $(a,b,c) \in \mathbb{R}^3$. We need to show that (a,b,c) can be expressed as a linear combination of the vectors in the set $\mathcal{B}$. Suppose

$$\begin{aligned}(a,b,c) &= \alpha_1(1,1,1) + \alpha_2(0,1,1) + \alpha_3(0,0,1)\\ &= (\alpha_1, \alpha_1+\alpha_2, \alpha_1+\alpha_2+\alpha_3)\end{aligned}$$

  This implies $\alpha_1 = a$, $\alpha_2 = b - a$ and $\alpha_3 = c - b - a$. Since we have obtained the coefficients, the linear combination is possible.

Thus, $\mathcal{B}$ is a basis for the vector space $\mathbb{R}^3$.

Example 4. Verify if $1 + x^2, x + x^3, 1 - x^2, x^3$ forms a basis for $\mathcal{P}_3$.

- First, LD or LI. Form the linear combination

$$a_1(1 + x^2) + a_2(x + x^3) + a_3(1 - x^2) + a_4(x^3) = \Theta = 0 + 0x + 0x^2 + 0x^3$$

  to get the system of equation

$$a_1 + a_3 = 0, \qquad a_2 = 0, \qquad a_1 - a_3 = 0, \qquad a_2 + a_4 = 0$$

  which gives $a_1 = a_2 = a_3 = a_4 = 0$. Hence the vectors are LI.

- Second, span. Let $p_0 + p_1x + p_2x^2 + p_3x^3$ be any polynomial in $\mathcal{P}_3$. To see if it is a linear combination of the given vectors, we consider the equation

$$p_0 + p_1x + p_2x^2 + p_3x^3 = \alpha_1(1 + x^2) + \alpha_2(x + x^3) + \alpha_3(1 - x^2) + \alpha_4(x^3).$$

  If we can find the $\alpha_i's$ (they will be in terms of the $p_i's$) then the span of the given vectors is $\mathcal{P}_3$ and hence $\mathcal{B} = \{1 + x^2, x + x^3, 1 - x^2, x^3\}$ is a basis for $\mathcal{P}^3$.

Can you prove that $\mathcal{B} = \{1, x, x^2, x^3\}$ is also a basis for $\mathcal{P}_3$?

Note 1. Let us consolidate the idea of a basis.

- A basis $\mathcal{B}$ contains the minimum number of vectors required to span a vector space. It means if we remove any vector from $\mathcal{B}$ it will not span the entire vector space (it will still span but a proper subspace of the vector space). Hence a basis is the ***minimal*** *spanning set of vectors of a vector space.*
- If we add any vector v to a basis $\mathcal{B} = \{v_1, v_2, \cdots, v_n\}$ of a vector space V it becomes a LD set. It is because v is a vector in V and hence is a linear combination of the basis vectors $v_i's$. Thus $\{v_1, v_2, \cdots, v_n, v\}$ is LD. This tells us that any basis contains the maximum number of vectors that can be LI. Hence a basis is also the ***maximal*** *linearly independent set of vectors in a vector space.*
- As seen from the examples, there can be many sets that are a basis of a vector space. But because of the above two properties, the number of vectors in all the basis sets remains the same. (Verify this from the examples!). The number of basis vectors is important and has a name which is the next topic.

0.1 Dimension

We often say the plane $\mathbb{R}^2$ is two-dimensional or the space $\mathbb{R}^3$ is three dimensional. This is due to the number of degrees of freedom that we require to describe a point in $\mathbb{R}^2$ or $\mathbb{R}^3$. We discuss the role of basis in determining the degrees of freedom.

Consider $\mathbb{R}^2$ in which it can be proved that the set $\{(1,0),(0,1)\}$ is a basis. Then the vector $(-2,6) = -2(1,0) + 6(0,1)$. We interpret this as the point $(-2,6)$ requiring two degrees of freedom to describe it:-2 units in the direction of the vector $(1,0)$ and 6 units in the direction of $(0,1)$.

Similarly, in the example above we have proved that $\{(1,1,1),(0,1,1),(0,0,1)\}$ is a basis for $\mathbb{R}^3$. The vector $(1,0,0)$, for example, can be expressed as $(1,0,0) = 1(1,1,1) - 1(0,1,1) + 0(0,0,1)$. Thus we have three degrees of freedom to describe the point $(1,0,0)$: it is 1 unit in the direction of the vector $(1,1,1)$, -1 unit in the direction of $(0,1,1)$ and 0 unit in the direction $(0,0,1)$.

The formal definition of dimension is as follows.

Definition of dimension

Definition 2. We say that a vector space is of **dimension** n if and only if it has a basis consisting of n elements.

The idea is the following. If a vector space V has a basis $\mathcal{B} = \{v_1, \cdots, v_n\}$, any vector v in V can be expressed uniquely as $v = \alpha_1 v_1 + \cdots \alpha_n v_n$. Then, we have exactly n degrees of freedom (*or* n *coordinates*) to describe the vector v uniquely.

Example 5. Suppose we are in $\mathbb{R}^3$. The dimension of this space is 3. Suppose we put the constraint that the z-coordinate is zero. That is, we are looking at points of the form $(x, y, 0)$. We can now choose only two co-ordinates freely. Thus, the dimension of the space in which this point can move is 2-dimensional: the xy-plane. Similarly, points of the form $(0, 0, z)$ can move only in a 1-dimensional space: the z-axis.

This example also tells us that there is a 1-dimensional *sub*-space inside a 2-dimensional *sub*-space inside a 3-dimensional space.

Example 6. Find the dimension of the vector space $\mathcal{P}_4$ consisting of all polynomials of degree upto 4.

We can verify that $\mathcal{B} = \{1, x, x^2, x^3, x^4\}$ is a basis of $\mathcal{P}_4$. Any polynomial, for example, $p(x) = 6 + 8x + 2x^4$ can be written as the linear combination $p(x) = 6(1) + 8(x) + 0(x^3) + 0(x^4) + 2(x^4)$ of the basis elements. We need 5 co-ordinates: $6, 8, 0, 0, 2$ to describe $p(x)$ uniquely. Hence dimension of $\mathcal{P}_4$ is 5.

Now, we put the constraint that the constant term is zero. Any polynomial in this space is of the form $ax^4+bx^3+cx^2+dx$. We can choose freely 4 real numbers to describe any vector in this space. Also a basis for this space is $\mathcal{B} = \{x, x^2, x^3, x^4\}$. Hence the dimension of this space is 4.

In general, the dimension of $\mathcal{P}_n$ is $n + 1$ as we need $n + 1$ coordinates to describe any polynomial in $\mathcal{P}_n$: 1 coordinate for the constant term and n coordinates for the $x, x^2, \cdots, x^n$ terms. Those coordinates appear as coefficients of the polynomial. If we add any constraint(s), like the constant term is 1 or the coefficient of x is 0 and so on, the dimension decreases.

Exercises

1. Find the dimension and a basis for the solution space of $Ax = 0$ and $Bx = 0$ where

$$A = \begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 4 & 1 & 3 \\ 3 & 6 & 1 & 4 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 2 & 0 & 2 & 1 \\ 3 & 6 & 1 & 9 & 6 \\ 2 & 4 & 1 & 7 & 5 \end{pmatrix}$$

   $\mathcal{B}_{Nullsp(A)} = \{(-2, 1, 0, 0), (-1, 0, -1, 1)\}$
   $\mathcal{B}_{Nullsp(B)} = \{(-2, 1, 0, 0, 0), (-2, 0, -3, 1, 0), (-1, 0, -3, 0, 1)\}$

2. Determine whether or not the set

$$B = \{(2, 3, 2), (1, 1, -1)\}$$

   is a basis for the space spanned by the set

$$S = \{(1, 2, 3), (5, 8, 7), (3, 4, 1)\}$$